

Ricardo A. Sanchez

Madison, WI 53705 | 787-590-9594

ricardo@sanchezsantiago.com

github.com/Ricardo-Sanchez-0

Professional Summary

Medical device engineer with 4+ years of hands-on implantable-device development spanning electrode-tissue interface characterization, charge injection limits, biopotential acquisition, stimulation safety, bench test fixture design, and regulated manufacturing. Experience developing and validating test methods, authoring test protocols and reports, maintaining requirements traceability and design history file documentation, and executing design verification and validation. Co-inventor on a pending neuromodulation patent application with transferable experience in electrophysiology systems. Biomedical engineering PhD completion is flexible to support full-time employment through early graduation or online completion.

Education

University of Wisconsin – Madison

Madison, WI

- *Doctor of Philosophy in Biomedical Engineering (early or online completion available)*

Expected Dec. 2027

Master of Science in Biomedical Engineering

Aug. 2024

University of Puerto Rico – Mayagüez

Mayagüez, PR

Bachelor of Science in Mechanical Engineering (Cum Laude)

Dec. 2023

Experience

University of Wisconsin – Ludwig Lab, Department of Biomedical Engineering &

Madison, WI

- **Neurosurgery**

Researcher (Advisor: Dr. Kip Ludwig)

Jun. 2022 – Present

- **Project: Electrochemical Safety Standards for Neuromodulation** – Bench and in vivo characterization of implantable electrodes to assess compliance with and establish charge injection limits and safety design guidelines aligned with regulatory standards for the neuromodulation field.
- Performed electrochemical impedance spectroscopy (EIS), voltage transient (VT) analysis, and cyclic voltammetry (CV) using actual neural waveforms to evaluate electrode design data against established safety thresholds; findings directly inform safety standards with broad implications for artificial electroactive surface area.
- Developed a quantitative framework relating electrode geometry to charge injection efficiency; established safety design guidelines identifying pulse-width-dependent transitions between geometric and electrochemical surface area-limited regimes.
- **Project: PS-OCT Guided Injectrode System** – Translational device development program targeting posterior tibial nerve stimulation (PTNS) for overactive bladder (OAB); advancing a minimally invasive electrode delivery platform from bench concept toward a defined clinical pathway.
- Designed and validated a PS-OCT-guided needle probe and bench test fixtures for percutaneous electrode and drug delivery; developed and validated test methods against functional requirements and authored test protocols and reports supporting requirements traceability and design history file documentation.
- Conducted ($n = 11$) large animal (porcine) experiments; acquired and analyzed EMG and ECAP biopotential data to quantify on- and off-target stimulation as a function of implantation depth. Characterized relevant anatomy with histological imaging.
- Developed a 3D computational modeling pipeline integrating microCT, MUSE fluorescence imaging, and histological segmentation to produce subject-specific anatomical models that guide electrode placement strategy and inform design iteration.
- Coordinated across imaging, surgical, and signal analysis teams, translating preclinical findings into device requirements, traceable design updates, and verification evidence aligned with patient safety objectives.

University of Wisconsin – Department of Biomedical Engineering

Madison, WI

Teaching Assistant, Graduate Seminar: Biomedical Startup Evaluation

Jan. 2026 – May 2026

- Co-instructed parts of a graduate-level course in which cross-functional student teams evaluated 12 biomedical startups spanning gene editing, neuromodulation, and orthopedic devices; applying structured biodesign frameworks to assess both technical feasibility and commercial viability.
- Taught applied frameworks for interpreting IP landscapes, FDA regulatory filings, design controls, and technology transfer strategy; equipped students to navigate the commercialization pathway from academic discovery to clinical impact.

- Facilitated venture capital pitch evaluations and hazard analyses, training technical researchers to identify high-risk translation barriers and communicate across the research-to-business interface.

• University of Puerto Rico–Mayagüez – Senior Capstone

Mayagüez, PR

4Front Stake Cable to Fitting Project (sponsored by Boston Scientific, Dorado, PR)

Jan. 2022 – Jun. 2022

- **Project: Manufacturing Process Redesign for RELIANCE 4-FRONT™ Lead** – Process improvement initiative to reduce defect rates and increase throughput on a high-volume ISO 13485-regulated medical device manufacturing line.
- Designed, prototyped, and deployed a motorized blade bench-test fixture to replace manual cutting operations on an ISO 13485-regulated line, eliminating operator-dependent variability.
- Implemented in-line fault detection sensors for real-time process deviation identification; authored design drawings, test protocols, verification reports, and work instructions supporting requirements traceability and manufacturing transfer.
- Validated process improvements through structured verification and validation (V&V) protocols and cross-functional review with quality and manufacturing engineers; demonstrated ~50% reduction in line dead time and \$422,328 in estimated annual cost savings.

Publications & Patents

• Patents

- Ludwig K, Trevathan J, Chen RCH, **Sanchez R.** Apparatus for applying therapy to a neurovascular bundle. US Patent App. 18/828,019. 2026.

• Journal Articles & Preprints

- Gholston AK, **Sanchez R.**, et al. Systematic errors in DC offset measurement and mitigation at neurostimulation electrodes: causes, implications and solutions. bioRxiv. 2025. doi:10.1101/2025.12.08.691908
- Deshmukh A, Cheng K, Settell ML, Knudsen BE, LaLuzerne M, Savage J, **Sanchez R.**, et al. Identification of low-threshold off-target activation pathways during stimulation of carotid baroreceptor afferents in swine. Journal of Neural Engineering. 2026;23(3):036022.

• Conference Abstracts

- **Sanchez R.**, et al. Decoding on- and off-target circuit recruitment during percutaneous tibial nerve stimulation for overactive bladder. NYC Neuromodulation Conference, New York, NY. 2026.
- Deshmukh A, Cheng K, **Sanchez R.**, et al. Identifying off-target activation pathways for Baroreflex Activation Therapy (BAT) in swine. Neuromodulation. 2024;27(7):S140–S141.

Skills

- **Regulatory & Standards:** ISO 13485, FDA guidance, Design Verification & Validation (V&V), Design Controls, Requirements Traceability, Design History File (DHF) Documentation, Test Protocol and Report Authoring, Standards Compliance, FDA-regulated preclinical protocols
- **Lab & Technical:** Test Method Development and Validation, Bench Test Fixture Design, Electrode–Tissue Interface Characterization, Electrochemical Analysis (EIS, CV, VT), Biopotential and EMG/ECAP Acquisition, Large Animal Models, Histological and OCT Imaging, 3D Printing, CNC Machining, PCB Design, Risk Analysis, Fabrication & Prototyping
- **Entrepreneurship & Translation:** Biodesign Process, Technology Transfer Strategy, IP Landscape Assessment, Venture Capital Evaluation, Neurotechnology Commercialization, Regulatory Strategy (FDA)
- **Computer / CAD:** COMSOL, Python, MATLAB, Fusion 360, OnShape, Siemens NX, Julia, Blender, Autodesk Eagle, Synapse, Ascent, NEURON, PyFibers
- **Language:** English (fluent), Spanish (native)